Partial-Parameter SAC: Complete Class-A, B, C Formulation for All Arrhenius Parameter Subsets

Extension of MUQ-SAC to Sparse Parameter Selection

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1 Scope and Notation

The modified Arrhenius log-rate-constant is written

$$\kappa(T) \;=\; \boldsymbol{\theta}(T)^\top \mathbf{p}, \qquad \boldsymbol{\theta}(T) = \begin{bmatrix} 1 \\ \ln T \\ -1/T \end{bmatrix}, \quad \mathbf{p} = \begin{bmatrix} \alpha \\ n \\ \varepsilon \end{bmatrix}, \tag{1}$$

where $\alpha = \ln A$, n is the temperature exponent, and $\varepsilon = E_a/R$. The full 3×3 Cholesky matrix $\mathbf{L}$ (lower triangular) satisfies $\Sigma_{\mathbf{p}} = \mathbf{L}\mathbf{L}^\top$ and parametrises samples as $\mathbf{p} = \mathbf{p}_0 + \mathbf{L}\boldsymbol{\zeta}$, $\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_3)$.

A *selection set* $S \subseteq \{\alpha, n, \varepsilon\}$ of size $m = |S|$ identifies the active parameters for a given reaction. We encode S by an index set $\mathcal{I} \subset \{0, 1, 2\}$ and:

- $\boldsymbol{\theta}_S(T)$: the m-vector formed by rows $\mathcal{I}$ of $\boldsymbol{\theta}(T)$.
- $\Sigma_r = \Sigma[\mathcal{I}, \mathcal{I}]$: the $m \times m$ principal submatrix of $\Sigma = \mathbf{L}\mathbf{L}^\top$.
- $\mathbf{L}_r$: the lower-triangular Cholesky factor of Σ_r, i.e. $\Sigma_r = \mathbf{L}_r \mathbf{L}_r^\top$.

- $\boldsymbol{\zeta}_r \in \mathbb{R}^m$: the reduced perturbation vector.
- The *reduced rate-constant perturbation*: $\Delta\kappa_S(T) = \boldsymbol{\theta}_S(T)^\top \mathbf{L}_r \boldsymbol{\zeta}_r$.
- The *reduced prior-uncertainty function*: $f_{\text{prior},S}(T) = \|\mathbf{L}_r^\top \boldsymbol{\theta}_S(T)\|_2$.

Direct extraction of rows and columns from $\mathbf{L}$ does *not* give $\mathbf{L}_r$. One must reconstruct $\Sigma = \mathbf{L}\mathbf{L}^\top$, extract Σ_r, then re-Cholesky: $\mathbf{L}_r = \text{chol}(\Sigma_r)$.

Table 1 summarises the seven possible selection sets and the reduced basis vector in each case.

Table 1: Reduced basis vector $\boldsymbol{\theta}_S(T)$ for all seven subsets.

S	m	$\boldsymbol{\theta}_S(T)$	Perturbation shape
$\{\alpha\}$	1	$[1]$	Constant in T
$\{n\}$	1	$[\ln T]$	Increases with $\ln T$
$\{\varepsilon\}$	1	$[-1/T]$	Decreases in magnitude with T
$\{\alpha, n\}$	2	$[1,\ \ln T]^\top$	Constant + logarithmic
$\{\alpha, \varepsilon\}$	2	$[1,\ -1/T]^\top$	Constant + inverse-T
$\{n, \varepsilon\}$	2	$[\ln T,\ -1/T]^\top$	Logarithmic + inverse-T
$\{\alpha, n, \varepsilon\}$	3	$[1,\ \ln T,\ -1/T]^\top$	Full Arrhenius (original MUQ-SAC)

2 Which Curve Classes Are Feasible?

Recall the definitions:

Class-A $\Delta\kappa_S(T)$ does *not* change sign on $[T_{\min}, T_{\max}]$. The perturbed curve lies entirely above or below κ_0.

Class-B $\Delta\kappa_S(T)$ *crosses* κ_0 and is additionally *tangent* to one uncertainty limit at an interior temperature T_u.

Class-C $\Delta\kappa_S(T)$ crosses κ_0 via a linear interpolation ($f_c(T)$) of the two anchor-point deviations at $T_{\min}$ and $T_{\max}$.

For classes B and C to exist two conditions must both hold:

1. **Sign-change condition.** There must exist $\boldsymbol{\zeta}_r \in \mathbb{R}^m$ such that $\Delta\kappa_S(T) = \boldsymbol{\theta}_S(T)^\top \mathbf{L}_r \boldsymbol{\zeta}_r$ changes sign on $(T_{\min}, T_{\max})$.
2. **Degrees-of-freedom condition (class-B only).** The constrained optimisation for class-B has $m+1$ unknowns ($\boldsymbol{\zeta}_r \in \mathbb{R}^m$ and T_u) and four equality constraints. An exact solution requires $m \geq 3$. For $m = 2$ the system is overdetermined (three equality constraints enforced, one relaxed); for $m = 1$ class-B is infeasible.

For a single-parameter set ($m = 1$), $\boldsymbol{\theta}_S(T)$ is a scalar function. Over the combustion range $T \in [800, 3000]\,\text{K}$:

- $\theta_\alpha(T) = 1 > 0$ always $\Rightarrow$ no sign change.
- $\theta_n(T) = \ln T > 0$ for $T > 1\,\text{K}$ $\Rightarrow$ no sign change.
- $\theta_\varepsilon(T) = -1/T < 0$ always $\Rightarrow$ no sign change.

Therefore, classes B and C are impossible for any single-parameter selection.

For two-parameter sets ($m = 2$) the perturbation is $\Delta\kappa_S(T) = c_1\, g_1(T) + c_2\, g_2(T)$ (where c_1, c_2 depend on $\boldsymbol{\zeta}_r$ and $\mathbf{L}_r$). A crossover occurs when $c_1/c_2 = -g_2(T_u)/g_1(T_u)$ has a solution $T_u \in (T_{\min}, T_{\max})$:

- $\{\alpha, n\}$: ratio $= -\ln T_u$, range $(-8.0, -6.7)$ over $[800, 3000]\,\mathrm{K}$ $\Rightarrow$ crossover at $T_u = \exp(-c_1/c_2)$. **Feasible.**
- $\{\alpha, \varepsilon\}$: ratio $= T_u$, range $(800, 3000)\,\mathrm{K}$ $\Rightarrow$ crossover at $T_u = -c_2/c_1 \cdot (-1) = c_2/c_1$. **Feasible.**
- $\{n, \varepsilon\}$: ratio $= (1/T_u)/\ln T_u$, unique solution of $T_u \ln T_u = c_2/c_1$; solvable for $c_2/c_1 \in [5360, 24000]$. **Feasible.**

Table 2 summarises the feasibility of each class for all seven cases.

Table 2: Feasibility of curve classes for each active-parameter subset.

S	Class-A	Class-B	Class-C	Comment
$\{\alpha\}$	✓	×	×	Constant shift; reduces to A-factor design
$\{n\}$	✓	×	×	Monotone shift $\propto \ln T$; no crossover
$\{\varepsilon\}$	✓	×	×	Monotone shift $\propto 1/T$; no crossover
$\{\alpha, n\}$	✓	✓	✓	Crossover at $T_u = \exp(-c_1/c_2)$
$\{\alpha, \varepsilon\}$	✓	✓	✓	Crossover at $T_u = c_2/c_1$
$\{n, \varepsilon\}$	✓	✓	✓	Crossover satisfies $T_u \ln T_u = c_2/c_1$
$\{\alpha, n, \varepsilon\}$	✓	✓	✓	Original MUQ-SAC (unchanged)

3 Single-Parameter Cases

For $m = 1$, $\mathbf{L}_r = [\sigma_S]$ (a positive scalar) and $\zeta_r = \zeta_S \in \mathbb{R}$.

The Cholesky scalar is $\sigma_\alpha = L_{11}$ (directly from the diagonal of $\mathbf{L}$). The reduced basis function and uncertainty are:

$$\theta_\alpha(T) = 1, \qquad f_{\text{prior},S}(T) = L_{11} \quad \text{(constant)}. \tag{2}$$

The perturbation $\Delta\kappa_\alpha = L_{11}\zeta_\alpha$ is temperature-independent. The **class-A sampling problem** is simply:

$$\min_{\zeta_\alpha \in \mathbb{R}} \sum_i \big(\alpha_s \cdot L_{11} - L_{11}\zeta_\alpha\big)^2 \qquad \Rightarrow \qquad \zeta_\alpha = \pm\alpha_s, \tag{3}$$

where $\alpha_s \in [0, 1)$ is a uniform random draw. The resulting sample:

$$\kappa(T) = \kappa_0(T) \pm \alpha_s L_{11}. \qquad \text{(A-factor class-A)}$$

This is exactly the existing A-factor design and requires no further development.

The marginal variance of n is $\sigma_n^2 = L_{21}^2 + L_{22}^2$, giving:

$$\sigma_n = \sqrt{L_{21}^2 + L_{22}^2}, \qquad \theta_n(T) = \ln T, \qquad f_{\text{prior},S}(T) = \sigma_n \ln T. \tag{4}$$

Note that $f_{\text{prior},S}(T)$ is strictly positive and *increases monotonically* with T. The **class-A sampling problem** is:

$$\min_{\zeta_n \in \mathbb{R}} \sum_i \big(\alpha_s \, \sigma_n \ln T_i - \sigma_n \zeta_n \ln T_i\big)^2 \qquad \Rightarrow \qquad \zeta_n = \pm \alpha_s. \tag{5}$$

The class-A sample:

$$\kappa(T) = \kappa_0(T) \pm \alpha_s \, \sigma_n \, \ln T. \tag{n-only class-A}$$

Physical meaning. The perturbation grows with $\ln T$, meaning the rate constant deviates more at high temperatures. The curve shape is a tilted (in Arrhenius space) version of the nominal with fixed sign of deviation. No crossover of κ_0 is possible since $\ln T > 0$ throughout the combustion temperature range.

The marginal variance of ε is $\sigma_\varepsilon^2 = L_{31}^2 + L_{32}^2 + L_{33}^2$:

$$\sigma_\varepsilon = \sqrt{L_{31}^2 + L_{32}^2 + L_{33}^2}, \qquad \theta_\varepsilon(T) = -\frac{1}{T}, \qquad f_{\text{prior},S}(T) = \frac{\sigma_\varepsilon}{T}. \tag{6}$$

The uncertainty function $f_{\text{prior},S}(T) = \sigma_\varepsilon / T$ *decreases* with temperature, reflecting that activation-energy uncertainty matters most at low temperatures. The **class-A sampling problem**:

$$\min_{\zeta_\varepsilon \in \mathbb{R}} \sum_i \left(\alpha_s \frac{\sigma_\varepsilon}{T_i} - \sigma_\varepsilon \zeta_\varepsilon \cdot \frac{-1}{T_i} \cdot (-1) \right)^2 = \min \sum_i \sigma_\varepsilon^2 \frac{(\alpha_s - \zeta_\varepsilon)^2}{T_i^2} \;\Rightarrow\; \zeta_\varepsilon = \pm \alpha_s. \tag{7}$$

The class-A sample:

$$\kappa(T) = \kappa_0(T) - \alpha_s \frac{\sigma_\varepsilon}{T} \quad \text{or} \quad \kappa_0(T) + \alpha_s \frac{\sigma_\varepsilon}{T}. \tag{ε-only class-A}$$

Physical meaning. An increase in $\varepsilon = E_a/R$ lowers κ most at low temperatures (strong Arrhenius effect) and has diminishing impact at high T. The sign of the perturbation is fixed: $\theta_\varepsilon(T) = -1/T < 0$ always, so no crossover.

4 Two-Parameter Cases

For $m = 2$, write $\mathbf{L}_r = \left[\begin{smallmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{smallmatrix}\right]$ (the Cholesky factor of the 2×2 principal submatrix Σ_r). The perturbation is:

$$\Delta\kappa_S(T) = \boldsymbol{\theta}_S(T)^\top \mathbf{L}_r \boldsymbol{\zeta}_r = (l_{11}\zeta_1)\, g_1(T) + (l_{21}\zeta_1 + l_{22}\zeta_2)\, g_2(T), \tag{8}$$

where (g_1, g_2) are the two components of $\boldsymbol{\theta}_S$.

4.1 $S = \{\alpha, n\}$: Pre-exponential and temperature exponent

$$\boldsymbol{\theta}_S(T) = \begin{bmatrix} 1 \\ \ln T \end{bmatrix}, \quad \mathbf{L}_r = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix} \text{ with } \Sigma_r = \begin{bmatrix} \sigma_\alpha^2 & \sigma_{\alpha n} \\ \sigma_{\alpha n} & \sigma_n^2 \end{bmatrix}. \tag{9}$$

$$\Delta\kappa(T) = \underbrace{(l_{11}\zeta_1)}_{a} + (l_{21}\zeta_1 + l_{22}\zeta_2) \ln T \;\equiv\; a + b \ln T. \tag{10}$$

Sign-change condition. The combination $a + b \ln T$ changes sign at

$$T_u = e^{-a/b} \in (T_{\min}, T_{\max}) \quad \Leftrightarrow \quad -\frac{a}{b} \in (\ln T_{\min}, \ln T_{\max}). \tag{11}$$

The reduced prior uncertainty:

$$f_{\text{prior},S}(T) = \left\|\mathbf{L}_r^\top [1, \ln T]^\top\right\|_2 = \sqrt{(l_{11} + l_{21}\ln T)^2 + (l_{22}\ln T)^2}. \tag{12}$$

Class-A ($\{\alpha, n\}$):

$$\boxed{\min_{\boldsymbol{\zeta}_r \in \mathbb{R}^2} \sum_i \left(\alpha_s\, f_{\text{prior},S}(T_i) - \boldsymbol{\theta}_S(T_i)^\top \mathbf{L}_r \boldsymbol{\zeta}_r\right)^2.} \tag{A, α-n}$$

The resulting curve $\kappa_0(T) + a + b\ln T$ has fixed-sign deviation.

Class-B ($\{\alpha, n\}$): There are $m + 1 = 3$ unknowns (ζ_1, ζ_2, T_u) and four constraints. Enforce three as equalities and minimise the objective:

$$\boxed{\begin{aligned} &\min_{\zeta_1,\zeta_2,T_u} \sum_i \left(f_{\text{prior},S}(T_i) - \boldsymbol{\theta}_S(T_i)^\top \mathbf{L}_r \boldsymbol{\zeta}_r\right)^2 \\ &\text{s.t.} \quad r_1 f_{\text{prior},S}(T_{\min}) - \boldsymbol{\theta}_S(T_{\min})^\top \mathbf{L}_r \boldsymbol{\zeta}_r = 0, \\ &\qquad\; r_2 f_{\text{prior},S}(T_{\max}) - \boldsymbol{\theta}_S(T_{\max})^\top \mathbf{L}_r \boldsymbol{\zeta}_r = 0, \\ &\qquad\; r_3 f_{\text{prior},S}(T_u) - \boldsymbol{\theta}_S(T_u)^\top \mathbf{L}_r \boldsymbol{\zeta}_r = 0, \\ &\qquad\; \frac{d}{dT_u}\left[r_3 f_{\text{prior},S}(T_u)\right] - \frac{d}{dT_u}\left[\boldsymbol{\theta}_S(T_u)^\top \mathbf{L}_r \boldsymbol{\zeta}_r\right] = 0, \end{aligned}} \tag{B, α-n}$$

where $r_1, r_2 \in [-1, 1]$ random, $r_3 = -\operatorname{sign}(r_1)$. The derivative of $\Delta\kappa(T) = a + b\ln T$ with respect to T is b/T, so the tangency condition simplifies to $r_3\, f'_{\text{prior},S}(T_u) = b/T_u$.

Class-C ($\{\alpha, n\}$):

$$\boxed{f_c(T) = r_1 f_{\text{prior},S}(T_{\min}) + \frac{r_2 f_{\text{prior},S}(T_{\max}) - r_1 f_{\text{prior},S}(T_{\min})}{T_{\max} - T_{\min}}(T - T_{\min}); \quad \min_{\boldsymbol{\zeta}_r} \sum_i \left(f_c(T_i) - \boldsymbol{\theta}_S(T_i)^\top \mathbf{L}_r \boldsymbol{\zeta}_r\right)^2.} \tag{C, α-n}$$

4.2 $S = \{\alpha, \varepsilon\}$: Pre-exponential and activation energy

$$\boldsymbol{\theta}_S(T) = \begin{bmatrix} 1 \\ -1/T \end{bmatrix}, \quad \Sigma_r = \begin{bmatrix} \sigma_\alpha^2 & \sigma_{\alpha\varepsilon} \\ \sigma_{\alpha\varepsilon} & \sigma_\varepsilon^2 \end{bmatrix}. \tag{13}$$

$$\Delta\kappa(T) = a - \frac{b}{T}, \qquad a = l_{11}\zeta_1, \quad b = l_{21}\zeta_1 + l_{22}\zeta_2. \tag{14}$$

Sign-change condition. The combination $a - b/T$ changes sign at

$$T_u = \frac{b}{a} \in (T_{\min}, T_{\max}) \quad \Leftrightarrow \quad \frac{b}{a} \in (800, 3000)\,\text{K}. \tag{15}$$

Physical interpretation: the constant pre-exponential contribution a is cancelled by the activation-energy contribution b/T at the crossover temperature. This is the most common crossover mechanism in combustion Arrhenius kinetics.

The reduced prior uncertainty:

$$f_{\text{prior},S}(T) = \sqrt{\left(l_{11} - \frac{l_{21}}{T}\right)^2 + \frac{l_{22}^2}{T^2}}. \tag{16}$$

The tangency derivative is $d\Delta\kappa/dT = b/T^2$, and the class-B tangency condition becomes $r_3 f'_{\text{prior},S}(T_u) = b/T_u^2$.

Class-A ($\{\alpha, \varepsilon\}$):

$$\boxed{\min_{\boldsymbol{\zeta}_r \in \mathbb{R}^2} \sum_i \left(\alpha_s\, f_{\text{prior},S}(T_i) - \boldsymbol{\theta}_S(T_i)^\top \mathbf{L}_r \boldsymbol{\zeta}_r\right)^2.} \tag{A, α-ε}$$

Class-B ($\{\alpha, \varepsilon\}$): Same structure as (B,α-n) with $\boldsymbol{\theta}_S = [1, -1/T]^\top$ and $d\Delta\kappa/dT = b/T^2$.

$$\boxed{\begin{aligned} &\min_{\zeta_1,\zeta_2,T_u} \sum_i \big(f_{\text{prior},S}(T_i) - \boldsymbol{\theta}_S(T_i)^\top \mathbf{L}_r \boldsymbol{\zeta}_r\big)^2 \\ &\text{s.t.} \quad r_1 f_{\text{prior},S}(T_{\min}) - \Delta\kappa(T_{\min}) = 0, \quad r_2 f_{\text{prior},S}(T_{\max}) - \Delta\kappa(T_{\max}) = 0, \\ &\qquad r_3 f_{\text{prior},S}(T_u) - \Delta\kappa(T_u) = 0, \quad r_3 f'_{\text{prior},S}(T_u) - b/T_u^2 = 0. \end{aligned}} \tag{B, α-ε}$$

Class-C ($\{\alpha, \varepsilon\}$):

$$\boxed{f_c(T) = r_1 f_{\text{prior},S}(T_{\min}) + \frac{r_2 f_{\text{prior},S}(T_{\max}) - r_1 f_{\text{prior},S}(T_{\min})}{T_{\max} - T_{\min}}(T - T_{\min}); \quad \min_{\zeta_r} \sum_i \big(f_c(T_i) - \Delta\kappa(T_i)\big)^2.} \tag{C, α-ε}$$

4.3 $S = \{n, \varepsilon\}$: Temperature exponent and activation energy

This case is *qualitatively different* from the two pairs involving α because neither $\ln T$ nor $-1/T$ alone can produce a crossover, yet the *combination* can.

$$\boldsymbol{\theta}_S(T) = \begin{bmatrix} \ln T \\ -1/T \end{bmatrix}, \quad \Sigma_r = \begin{bmatrix} \sigma_n^2 & \sigma_{n\varepsilon} \\ \sigma_{n\varepsilon} & \sigma_\varepsilon^2 \end{bmatrix}. \tag{17}$$

With $\mathbf{L}_r = \left[\begin{smallmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{smallmatrix}\right]$:

$$\Delta\kappa(T) = (l_{11}\zeta_1)\ln T - \frac{l_{21}\zeta_1 + l_{22}\zeta_2}{T} \;\equiv\; a \ln T - \frac{b}{T}. \tag{18}$$

Here $a = l_{11}\zeta_1$ and $b = l_{21}\zeta_1 + l_{22}\zeta_2$.

Sign-change condition. $\Delta\kappa(T_u) = 0$ requires

$$a \ln T_u = \frac{b}{T_u} \quad \Leftrightarrow \quad T_u \ln T_u = \frac{b}{a}. \tag{19}$$

The function $h(T) = T \ln T$ is strictly increasing for $T > 1/e$ (always true in combustion), so there is a *unique* crossover temperature T_u for each value of $b/a > 0$. The crossover exists in $(T_{\min}, T_{\max})$ when

$$\frac{b}{a} \in \big(T_{\min} \ln T_{\min},\ T_{\max} \ln T_{\max}\big) \approx \big(5358,\ 24000\big)\,\text{K} \quad (\text{for } [800, 3000]\,\text{K}). \tag{20}$$

Physical meaning. This is the kinetic *compensation effect*: increasing both n and ε together means the rate constant increases at high T (the n effect) but decreases at low T (the E_a effect), with a unique crossover temperature. The fact that neither parameter individually can produce this behaviour, yet their combination can, is a direct consequence of their correlation encoded in Σ_r.

The reduced prior uncertainty:

$$f_{\text{prior},S}(T) = \sqrt{(l_{11} \ln T - l_{21}/T)^2 + (l_{22}/T)^2}. \tag{21}$$

The derivative $d\Delta\kappa/dT = a/T + b/T^2$, and the class-B tangency condition is: $r_3 f'_{\text{prior},S}(T_u) = a/T_u + b/T_u^2$.

Class-A ($\{n, \varepsilon\}$):

$$\boxed{\min_{\zeta_r \in \mathbb{R}^2} \sum_i \big(\alpha_s\, f_{\text{prior},S}(T_i) - \Delta\kappa(T_i)\big)^2, \qquad \Delta\kappa(T) = a \ln T - b/T.} \tag{A, n-ε}$$

Class-B ($\{n, \varepsilon\}$):

$$\boxed{\begin{aligned} &\min_{\zeta_1,\zeta_2,T_u} \sum_i \big(f_{\text{prior},S}(T_i) - \Delta\kappa(T_i)\big)^2 \\ &\text{s.t.} \quad r_1 f_{\text{prior},S}(T_{\min}) - \Delta\kappa(T_{\min}) = 0, \quad r_2 f_{\text{prior},S}(T_{\max}) - \Delta\kappa(T_{\max}) = 0, \\ &\qquad r_3 f_{\text{prior},S}(T_u) - \Delta\kappa(T_u) = 0, \quad r_3 f'_{\text{prior},S}(T_u) - (a/T_u + b/T_u^2) = 0. \end{aligned}} \tag{B, n-ε}$$

The class-B problem has $m + 1 = 3$ unknowns (ζ_1, ζ_2, T_u) but 4 constraints. The system is overdetermined. In practice, the SLSQP solver will minimise the sum of squared constraint violations together with the objective. Alternatively, enforce constraints 1, 3, and 4 as strict equalities and treat constraint 2 as a soft penalty. The solution will be approximate but remains within the uncertainty domain. This is in contrast to the full $m = 3$ case, which has exactly 4 unknowns and 4 constraints.

Class-C ($\{n, \varepsilon\}$):

$$\boxed{f_c(T) = r_1 f_{\text{prior},S}(T_{\min}) + \frac{r_2 f_{\text{prior},S}(T_{\max}) - r_1 f_{\text{prior},S}(T_{\min})}{T_{\max} - T_{\min}}(T - T_{\min}); \quad \min_{\boldsymbol{\zeta}_r} \sum_i \big(f_c(T_i) - \Delta\kappa(T_i)\big)^2.} \tag{C, n-ε}$$

5 Full Three-Parameter Case: Recap

This is the original MUQ-SAC formulation. For $S = \{\alpha, n, \varepsilon\}$, $m = 3$, $\mathbf{L}_r = \mathbf{L}$ (the full Cholesky matrix), and $\boldsymbol{\theta}_S = \boldsymbol{\theta}$. The class-B problem has exactly $3 + 1 = 4$ unknowns and 4 constraints, giving an (approximately) exact solution. All formulas reduce to those in the original paper [1] and are not repeated here.

6 Degrees of Freedom and Constraint Count

Table 3 summarises the constraint structure. The deficit $\delta = 4 - (m + 1)$ measures how overdetermined class-B is; $\delta = 0$ is exactly determined.

Table 3: Constraint structure for class-B optimisation.

S	m	Unknowns $(m+1)$	Constraints	Deficit δ	Feasibility
$\{\alpha\}$,$\{n\}$,$\{\varepsilon\}$	1	2	4	2	Class-B impossible
$\{\alpha, n\}$,$\{\alpha, \varepsilon\}$,$\{n, \varepsilon\}$	2	3	4	1	Overdetermined (soft)
$\{\alpha, n, \varepsilon\}$	3	4	4	0	Exactly determined

7 Shape of Class-A Curves Across All Cases

Even though class-A curves always stay on one side of κ_0, their *temperature dependence* varies qualitatively. Table 4 captures this.

Table 4: Temperature dependence of the class-A curve perturbation $\Delta\kappa_S(T)$.

S	$\Delta\kappa_S(T)$ (class-A)	Limiting behaviour
$\{\alpha\}$	$\pm\alpha_s L_{11}$	Flat; same deviation at all T
$\{n\}$	$\pm\alpha_s\sigma_n \ln T$	Grows with T; small at low T
$\{\varepsilon\}$	$\mp\alpha_s\sigma_\varepsilon/T$	Decays with T; large at low T
$\{\alpha, n\}$	$a + b\ln T$ (fixed sign)	Mixed: flat+log
$\{\alpha, \varepsilon\}$	$a - b/T$ (fixed sign)	Mixed: flat−decaying
$\{n, \varepsilon\}$	$a\ln T - b/T$ (fixed sign)	Log-growth minus $1/T$-decay
$\{\alpha, n, \varepsilon\}$	Full Arrhenius shape	Original MUQ-SAC

A $\{n\}$-only class-A curve has its largest deviation at *high* temperatures (where $\ln T$ is largest), while a $\{\varepsilon\}$-only class-A curve has its largest deviation at *low* temperatures (where $1/T$ is largest). This is physically meaningful: temperature-exponent uncertainty dominates at high T while activation-energy uncertainty dominates at low T.

8 Canonical Crossover Temperatures

For completeness, Table 5 gives the analytical crossover condition for each two-parameter case, where a and b are defined by (10), (14), and (18) respectively.

Table 5: Crossover temperature T_u for two-parameter cases.

S	Crossover condition	Analytic T_u	Existence condition
$\{\alpha, n\}$	$a + b\ln T_u = 0$	$T_u = e^{-a/b}$	$-a/b \in (\ln T_{\min}, \ln T_{\max})$
$\{\alpha, \varepsilon\}$	$a - b/T_u = 0$	$T_u = b/a$	$b/a \in (T_{\min}, T_{\max})$
$\{n, \varepsilon\}$	$a\ln T_u = b/T_u$	$T_u \ln T_u = b/a$ (implicit)	$b/a \in (T_{\min}\ln T_{\min}, T_{\max}\ln T_{\max})$

9 Summary: Reduced Sampling Operators

The complete mapping from selection set S to the sampling operators is given in Table 6.

Table 6: Complete operator set for all seven active-parameter selections.

S	$\boldsymbol{\theta}_S(T)$		Classes	Comment
$\{\alpha\}$	$[1]$	$[L_{11}]$	A only	Existing A-factor design
$\{n\}$	$[\ln T]$	$[\sigma_n]$	A only	Growing deviation
$\{\varepsilon\}$	$[-1/T]$	$[\sigma_\varepsilon]$	A only	Decaying deviation
$\{\alpha, n\}$	$[1, \ln T]^\top$	$\mathrm{chol}(\Sigma_{\alpha n})$	A, B, C	Crossover at $e^{-a/b}$
$\{\alpha, \varepsilon\}$	$[1, -1/T]^\top$	$\mathrm{chol}(\Sigma_{\alpha\varepsilon})$	A, B, C	Crossover at b/a
$\{n, \varepsilon\}$	$[\ln T, -1/T]^\top$	$\mathrm{chol}(\Sigma_{n\varepsilon})$	A, B, C	Crossover: $T_u \ln T_u = b/a$
$\{\alpha, n, \varepsilon\}$	$[1, \ln T, -1/T]^\top$	$\mathbf{L}$	A, B, C	Original MUQ-SAC

10 Consistency of the Reduced Samples

For any selection S with principal submatrix Σ_r and Cholesky factor , the reduced samples $\mathbf{p}_S = \mathbf{p}_{0,S} + \boldsymbol{\zeta}_r$ satisfy:

$$\mathbf{p}_S \sim \mathcal{N}(\mathbf{p}_{0,S},\ \Sigma_r), \tag{22}$$

which is the correct *marginal* distribution of the selected parameters under the full model $\mathbf{p} \sim \mathcal{N}(\mathbf{p}_0, \Sigma)$.

The reduced uncertainty ellipsoid $\mathcal{E}_{\Sigma_r} \subseteq \mathcal{E}_\Sigma$ (proved via the principal submatrix theorem: all eigenvalues of Σ_r are dominated by those of Σ via Cauchy interlacing). Equivalently, $f_{\text{prior},S}(T) \leq f_{\text{prior}}(T)$ for all T, so no reduced sample ever exceeds the original uncertainty bounds.

For the single-parameter cases the uncertainty function has a simple factored form:

$$f_{\text{prior},S}[\alpha](T) = L_{11}, \quad f_{\text{prior},S}[n](T) = \sigma_n \ln T, \quad f_{\text{prior},S}[\varepsilon](T) = \sigma_\varepsilon / T, \tag{23}$$

all strictly less than $f_{\text{prior}}(T) = \|\mathbf{L}^\top \boldsymbol{\theta}(T)\|_2$ as expected.

The reduced covariance $\Sigma_r = \Sigma_{SS}$ (principal submatrix, marginal) is used rather than the conditional covariance $\Sigma_{\text{cond}} = \Sigma_{SS} - \Sigma_{S\bar{S}}\Sigma_{\bar{S}\bar{S}}^{-1}\Sigma_{\bar{S}S}$ (which would hold if inactive parameters were fixed at specific non-nominal values). Since $\Sigma_{\text{cond}} \preceq \Sigma_{SS}$ (Schur complement), the marginal choice is conservative and generates a larger, more robust sample space. It does not assume any knowledge of where the inactive parameters will be set during optimisation.

References

[1] K. Panchal, V. Vasudevan, S. Ambikasaran, K. Narayanaswamy, "A novel uncertainty quantification methodology and sampling technique for joint probability distribution of the Arrhenius rate expression," *submitted*, 2024.